

Expert speak

C.P. Ravikumar, technical director-unity relation, Texas Instruments (TI) India, spoke about the gap between industry requirements and candidate skill sets, selection criteria for fresh recruits and need of specialised knowledge training for aspirants. Excerpts:



C.P. Ravikumar

Do you feel that there is a gap between industry requirements and candidate skill sets?

We do see a gap in areas such as VLSI design and embedded systems software, which are TI India's focus areas. In particular, getting people in analogue design is a challenge.

What skills are required?

The skill sets are quite varied. I can speak on behalf of TI India, which mainly focuses on VLSI design and embedded software. VLSI design is an area where progresses are rapid.

There are two main aspects in VLSI design, namely, front-end and back-end. Front-end design includes digital design using hardware description languages, design verification through simulation and formal verification techniques, synthesis of digital designs to gates, and design for testability. Since SoC (system-on-chip) design today happens through a process of integration of existing IP blocks, the engineers must have a good knowledge of computer architecture, different IPs such as DMA controllers and USB controllers, and bus protocols such as OCP.

The VLSI design engineer must have exposure to fundamental concepts as well as familiarity with tools and design flows. Universities, on the contrary, tend to provide an overview of all the aspects of VLSI design. This is one reason why managers may complain that engineers are not production-ready. Having more MS and Ph.D students should help to alleviate this problem.

Further, to practise VLSI design, which is mainly orchestrated through automated flows, the engineer must know quite a bit of UNIX utilities and productivity-enhancement tools such as Perl.

Analogue design is a competency which requires a deep understanding of mathematical concepts of circuit design, network analysis and control theory. Getting campus-hires for analogue

design is a big challenge.

Back-end design consists of CMOS library design and characterisation, physical design (floor-planning, placement and routing), design for manufacturability, packaging, test generation and fault simulation. A good understanding of power dissipation in circuits is important from the viewpoint of controlling the dynamic and static power dissipation. Issues such as on-chip variability of transistor parameters and design for manufacturability are becoming important in modern CMOS technologies.

What's the selection criteria for fresh recruits?

The subject of VLSI design is quite vast. It is not fair to expect both breadth and depth from a fresh recruit. At the undergraduate level, we expect a good understanding of fundamental concepts. Our campus recruitment is based on a written test and an interview.

The written test is separate for hardware and software engineers. The hardware paper has a section on aptitude, a section on digital design and a section on analogue design—we ask the students to select any one section between digital and analogue design. The software paper has two parts—an aptitude section and a section on software. Candidates who fare well in the written test are shortlisted for interview.

From my experience, the interviews are mostly based on problem-solving exercises and test the conceptual understanding rather than rote learning. We don't pay too much importance to the B.Tech project done by the student.

For postgraduate students, our process is similar, except that we may go into the M.Tech project/MS thesis/Ph.D thesis. For candidates with a Ph.D background, we also ask for a presentation where we invite our experts.

How important is specialised knowledge training for aspirants?

Specialised knowledge is useful for the engineers to grow inside the company. More than the knowledge itself, the ability to learn is a more important aspect. This learning can happen from peers, from publications, from attending meetings and conferences, etc. Many of our engineers express their desire to take up higher studies and this is encouraged.

learners. The main job functions are engineering aspects of design, product, test, applications and process. Product engineering and test engineering functions are often combined efficiently into one role because of the interdependency and overlap of skills and tasks. Though most of the entry-level job profiles involve code entry, verification, synthesis and system-level engineering works, the possibility of role diversification moving ahead is huge.

The roles are defined depending

on the requirements of the job. Typically, a design engineer takes specifications, defines architecture, designs circuits, runs simulations, supervises layout, tapes out the chip to the foundry and evaluates the prototype once the chip comes back from the factory. The design role itself offers further classification. For example, when the marketing team tells a company what product to design for the market, the architecture team is responsible for defining the chip design. You may work in such a team as a chip archi-

tect. As an RTL designer, you may be involved in designing the functions or modules of the chip which are required to perform the well-defined tasks described by the architecture team. This design is written in a high-level logic description language such as Verilog or VHDL.

If you find interest in optimising the existing standard cell library for performance and/or power by optimising circuit design or design/layout of custom cells like adders and shifters, synthesising the logic description